21 . Spirally traversing a matrix

```
class Solution {
  public:
    vector<int> spirallyTraverse(vector<vector<int>> &mat) {
        vector<int> ans;
        int n = mat.size();
        if (n == 0) return ans;
        int m = mat[0].size();
        int top = 0, bottom = n - 1;
        int left = 0, right = m - 1;
        while (top <= bottom && left <= right) {
            // 1) left -> right along top row
            for (int j = left; j <= right; ++j) ans.push_back(mat[top][j]);
            ++top;     // 2) top -> bottom along right column
            for (int i = top; i <= bottom && left <= right; ++i) ans.push_back(mat[i][right]);
            --right;       // 3) right -> left along bottom row (if still valid)
            if (top <= bottom) {
                for (int j = right; j >= left; --j) ans.push_back(mat[bottom][j]);
                --bottom;
            }   // 4) bottom -> top along left column (if still valid)
            if (left <= right) {
                for (int i = bottom; i >= top; --i) ans.push_back(mat[i][left]);
                ++left;
            }
        }       return ans;
    }
```